FLAVIO D. GUALTIERI

flavio-gualtieri • f.d.gualtieri@qmul.ac.uk • +44 7767 366389

EDUCATION

Queen Mary University of London — *Dec 2025 – Dec 2028*
Ph.D. in Mathematics • Supervisors: Omer Bobrowski & Primož Skraba • Funded by the Erlangen AI Hub — London, UK

Researching deep learning models for generation and inference on stochastic point processes.

King's College London — *Sep 2023 – Sep 2024*
M.Sc. in Theoretical Physics — London, UK

Coursework: quantum field theory, conformal field theory, Lie algebras, stochastic processes.
Dissertation: Topological Surface Codes on Exactly Solvable Lattice Models.

University of Chicago — *Sep 2019 – Jun 2023*
B.S. in Mathematics & B.A. in Physics — Chicago, USA

Coursework: abstract algebra, linear algebra, martingales and Brownian motion, statistical models and methods, statistical mechanics, advanced quantum mechanics, experimental physics.

RESEARCH & PROJECTS

Simulation-Based Inference for Spatial Point Processes — *Likelihood-Free Inference, Classification*

- From one observed point pattern, a classifier picks the generating family and a per-family neural estimator infers its parameters, trained on 900k simulations. Inputs combine summary functions and topological features.
- Designed evaluation method: simulate new patterns using the model's output and score against the input with a kernel score on local configurations, giving a score from 0 (random fit) to 1 (true model). The model scores 0.90, against 0.74 for classical baselines.

Eigengrokking: Can Hessian Curvature Forecast Grokking? — *Transformers, Training Dynamics*

- Trained 240 transformers on modular arithmetic, tracked Hessian and Gauss–Newton spectra through training, and forecast each run's grokking time from the first 500 steps after memorization. Built a stochastic Lanczos quadrature pipeline, validated against exact Hessians on smaller models.
- Found that curvature adds only a modest improvement over simple training signals (loss, weight and gradient norms) in explaining why identically configured runs grok at different times: R^2 increases 0.17→0.28 (ridge regression) and 0.48→0.60 (gradient boosted trees) on unseen configurations. The sharpest spectral signal is a collapse in effective rank, but it coincides with grokking rather than preceding it. Traced post-grokking accuracy crashes to AdamW's default $\beta_2 = 0.999$; applying $\beta_2 = 0.98$ cut them tenfold.

DeepGARCH: Feature-Conditioned Neural Volatility Model — *Volatility Forecasting, Time Series*

- Built a natural gas volatility forecaster in which a neural network (MLP) updates a GARCH(1,1) model's parameters each day from recent returns, realized volatility, volume and EIA storage reports, trained end to end. Over 10 training runs it had the lowest forecast error (QLIKE) of five models, ahead of GARCH and EGARCH, though not by a significant margin (Diebold–Mariano $p = 0.11$). The edge depends on tightly capping the model's long-run volatility level, which matters after the 2021 gas-price shock but cannot be detected on pre-2021 validation data.

EXPERIENCE

PwC Data & AI — *Dec 2024 – Dec 2025*
Data Scientist (Associate) — Milan, Italy

- Built an outage-detection pipeline, deployed on the client's AWS, processing 5 GB of telemetry per day. An XGBoost time-series model forecasts expected live user counts; a drop below the forecast flags a possible outage and triggers DBSCAN clustering on the locations of recently disconnected users. Cluster sizes are compared with each area's historical norm, with thresholds tuned by grid search, to separate genuine localized outages from background noise. Cut detection delay from 6 h to 30 min.
- Engineered new categorical and numerical features for a probabilistic deep-learning model forecasting shipment arrival times: a dense neural network (TensorFlow) outputs Weibull shape and scale parameters, trained with a negative log-likelihood loss, giving a full distribution over arrival times. Tuned hyperparameters with Optuna to 85% held-out coverage. Presented SHAP feature importances to the client for model explainability.

Center for RISC, University of Chicago — *Sep 2022 – May 2023*
Data Science Intern — Chicago, USA

- Built the declustering stage of a pipeline auditing a geospatial-intelligence firm's public earthquake predictions against live USGS data. Declustering reduces each seismic sequence to one 'mainshock', so predictions cannot score a hit on an aftershock by chance. Designed a rule-based clustering method: events are grouped only if *every* pair lies within 50 miles and 3 days, and the largest-magnitude event in each group is the mainshock.

SKILLS & LANGUAGES

Programming: Python [*PyTorch, TensorFlow, statsmodels, arch, XGBoost, pandas, NumPy, GUDHI*], SQL, R
Quantitative: machine learning, neural networks, diffusion models, gradient boosting, statistical inference, anomaly detection, time-series forecasting, Monte Carlo methods, stochastic point processes, topological data analysis
Compute: Git, AWS [*EC2, S3*], Slurm HPC
Languages: English & Italian [*native*]
Interests: world capitals